

Sparse estimator: For our sparse estimator, it is easy to see that for any $E_S \in \mathbb{R}^{n \times n}$ that has row sparsity k (each row has fewer than k non-zeros),

$$\|M - E_S\|_F \geq \Omega(\sqrt{n(n-k)}).$$

This implies that in order to achieve error $O(\sqrt{n})$, we would need $n - k = O(1)$, which requires $\Omega(n^2)$ parameters. $\square$

Now we construct a matrix that shows better separation between the approximation capability of sparse + low-rank vs sparse or low-rank alone.

Example 2. Consider the following randomized construction of matrix $Q \in \mathbb{R}^{n \times d}$ with $d \geq 6\epsilon^{-2}r \log n$ and $d = \Theta(\epsilon^{-2}r \log n)$ ($\epsilon \in (0, 1]$ and close to 0 and r is $\Theta(\log n)$): each entry of Q is picked independently and uniformly at random from $\{\pm\sqrt{r/d}\}$. Let $M = \sigma(QQ^\top)$ where σ is the elementwise exponential function.

Similar to Example 1, with high probability, we have:

$$(QQ^\top)_{i,j} = \begin{cases} r, & \text{if } i = j; \\ \in [-\epsilon, \epsilon], & \text{otherwise.} \end{cases}$$

We also have:

$$M_{i,j} = \begin{cases} e^r, & \text{if } i = j; \\ \in [1 - O(\epsilon), 1 + O(\epsilon)], & \text{otherwise.} \end{cases}$$

By setting r appropriately, we can formalize the separation between the approximation ability of sparse, low-rank, and sparse + low-rank matrices:

Theorem 4. Let M be the attention matrix from Example 2. Any sparse or low-rank estimator of M needs $\Omega(n^2)$ parameters for $O(n)$ error with probability at least $1 - n^{-1}$ while a sparse + low-rank estimator needs $O(n)$ parameters for $O(n)$ error with probability at least $1 - n^{-1}$.

Proof of Theorem 4. Similar to the proof of Theorem 3, by Hoeffding's inequality, for a pair $i \neq j$, we have that

$$\mathbb{P}(|q_i^\top q_j - \mathbb{E}[q_i^\top q_j]| \geq \epsilon) \leq 2 \exp\left(-\frac{2\epsilon^2}{\left(\frac{r}{\sqrt{d}} - \frac{-r}{\sqrt{d}}\right)^2}\right) = 2 \exp\left(-\frac{d\epsilon^2}{2r}\right).$$

Note that $\mathbb{E}[q_i^\top q_j] = 0$. By a union bound over all pairs $i \neq j$ (there are $n(n-1)/2$ such pairs), with probability at least $1 - n^{-1}$ (since $d \geq 6\epsilon^{-2}r \log n$), we have that

$$q_i^\top q_j \in [-\epsilon, \epsilon] \quad \text{for all } i \neq j.$$

Since we assume that $d \geq 6\epsilon^{-2} \log n$, we have that For the rest of the proof, we only consider this case (where $q_i^\top q_j \in [-\epsilon, \epsilon]$ for all $i \neq j$).

Let $T = M - (e^r - 1) \cdot I + J$, where J is the all one matrix. We see that T is zero on the diagonal. Moreover, using the fact that $e^x \leq 1 + 2|x|$ for all $x \in [-1, 1]$, the off-diagonal entries of T have of magnitude at most 2ϵ .

We consider 3 different estimators.

Sparse + low-rank estimator: Our estimator is

$$E_{SL} = \underbrace{(e^r - 1) \cdot I}_{\text{sparse}} + \underbrace{J}_{\text{low-rank}},$$

where $(e - 1)I$ has row sparsity 1 and $\text{rank}(J) = 1$.

The Frobenius error of sparse + low-rank approximation is

$$\|M - E_{\text{SL}}\|_F \leq O(\sqrt{\epsilon^2 n(n-1)}) \leq O(\epsilon n).$$

We have that:

- (i) Sparse + low-rank parameter count is $n \cdot (1 + 1) \leq O(n)$.
- (ii) Its Frobenius error is $\leq O(n)$.

Low-rank estimator: We want to argue that low-rank approximation would require more parameters. From a similar observation that any matrix R with rank that $n - \text{rank} = \Omega(1)$,

$$\|(e^r - 1)I - R\|_F \geq \Omega(e^r),$$

(by Eckart–Young–Mirsky theorem), we obtain a similar result to the proof of Theorem 3.

If R' is a matrix with rank such that $n - \text{rank} = \Omega(1)$, then $\|M - R'\|_F \geq \Omega(n) - \|T\|_F \geq \Omega(n) - O(\epsilon n) \geq \Omega(n)$. Hence any low-rank matrix with $O(n^2)$ parameters would have error $\Omega(n)$.

Sparse estimator: Similar to the proof of Theorem 3, for our sparse estimator, it is easy to see that for any $E_S \in \mathbb{R}^{n \times n}$ that has row sparsity k (each row has fewer than k non-zeros),

$$\|M - E_S\|_F \geq \Omega(\sqrt{n(n-k)}).$$

This implies that to get $O(n)$ error, we would need $\Omega(n^2)$ parameters. $\square$

D.2 Generative Model, Softmax Temperature, and Matrix Approximation

Here we show 3 cases where depending on the softmax temperature, either we'll need low-rank, low-rank + sparse, or sparse to approximate the attention matrix.

We start with some notation first. Given a matrix B , let $B[i, j]$ be the entry in the i th row and j th column. For a range $[l, r]$, we define a matrix $B_{[l, r]}$ where $B_{[l, r]}[i, j] = B[i, j]$ if $B[i, j] \in [l, r]$ and $B_{[l, r]} = 0$ otherwise (that is, $B_{[l, r]}$ only keep entries for B that are in the range $[l, r]$, with other entries zeroed out). We write $\text{supp}(C)$ for the set of locations of non-zeros in C . We let $\lambda_i(D)$ be the i -th largest (in absolute value) eigenvalue of D .

To prove Theorem 1, we first define a more general matrix class, prove that the attention matrix in Process 1 is a subset of this class (with high probability), and then show that Theorem 1 holds for this more general class. We introduce an extra parameter $l \in \mathbb{R}$, in addition to the inverse temperature β and the intro-cluster distance Δ .

Matrix Class 1. Let $Q \in \mathbb{R}^{n \times d}$ with every row of Q having ℓ_2 -norm in $[1 - O(\Delta), 1 + O(\Delta)]$, and let $A = QQ^\top$. Further:

1. Let $H = A_{[1/l, 2-1/l]}$ for some $l \geq \Omega(1)$. Assume that H is block diagonal with $\Omega(n)$ blocks, and $\text{supp}(H)$ is $o(n^2)$. That is, the large entries of $QQ^\top$ form a block diagonal matrix.
 2. Let $L = A - H$ then $L = A_{[-\Delta, \Delta]}$ where $\Delta = o(1/\log d)$. Assume that there is a constant fraction of elements in $\text{supp}(L)$ falling in $[0, \Delta]$. Assume that $\text{supp}(A_{[0, \Delta]})$ is $\Omega(n^2)$.
- Let $M_\beta = \exp(\beta \cdot A)$.

We now show that Process 1 is a subset of Matrix Class 1, with high probability.

Lemma 5. The matrix M_β in Process 1 is a subset of Matrix Class 1, where $l = \frac{1}{1-\Delta^2}$.

Proof. We first bound the norm of each row in Q in Process 1. For any i, j , we have

$$\|z_{ij}\|^2 = \|c_i + r_{ij}\|^2 = \|c_i\|^2 + 2c_i^\top r_{ij} + \|r_{ij}\|^2.$$

Since $c_i \sim \mathcal{N}(0, I_d/\sqrt{d})$, $\|c_i\|^2 \in [1 - \Delta^2, 1 + \Delta^2]$ with probability at least $1 - 2e^{-d\Delta^2/8}$ (by the standard argument using the fact that χ^2 -random variables are sub-exponential). Similarly, $\|r_{ij}\|^2 \in [\Delta^2 - \Delta^4, \Delta^2 + \Delta^4]$ with probability at least $1 - 2e^{-d\Delta^2/8}$. By concentration of measure, we can also bound $2c_i^\top r_{ij} \in [2\Delta - 2\Delta^3, 2\Delta + 2\Delta^3]$ as well. Therefore, we have that $\|z_{ij}\|^2 \in [1 - O(\Delta), 1 + O(\Delta)]$.

Now we show that the large entries of $QQ^\top$ form a block diagonal matrix. With high probability, the large entries come from intra-cluster dot product, and the small entries come from inter-cluster dot product.

We bound the intra-cluster dot product:

$$\begin{aligned} z_{ij}^\top z_{ik} &= (c_i + r_{ij})^\top (c_i + r_{ik}) \\ &= \|c_i\|^2 + c_i^\top r_{ij} + c_i^\top r_{ik} + r_{ij}^\top r_{ik}. \end{aligned}$$

Similar to the argument above, by concentration of measure, $\|c_i\|^2 \in [1 + \epsilon\Delta, 1 - \epsilon\Delta]$ with high probability (we will pick $\epsilon = \theta(\Delta)$). The cross terms $c_i^\top r_{ij}$ and $c_i^\top r_{ik}$ can be bounded using Cauchy-Schwarz inequality to be in $[-\epsilon\Delta, \epsilon\Delta]$ with high probability. And the fourth term $r_{ij}^\top r_{ik}$ is in $[-\epsilon\Delta^2, \epsilon\Delta^2]$ with high probability. Therefore, the inner product is in $1 \pm O(\epsilon\Delta)$ with high probability. This satisfies the first condition in Matrix Class 1, for $l = \frac{1}{1-\Delta^2}$, assuming $\epsilon \leq \Delta$.

We use a similar argument to bound the inter-cluster dot product. For $i \neq i'$

$$\begin{aligned} z_{ij}^\top z_{i'k} &= (c_i + r_{ij})^\top (c_{i'} + r_{i'k}) \\ &= c_i^\top c_{i'} + c_i^\top r_{i'k} + c_{i'}^\top r_{ij} + r_{ij}^\top r_{i'k}. \end{aligned}$$

By concentration of measure, $c_i^\top c_{i'} \in [-\epsilon, \epsilon]$. Similar to the argument in the intra-cluster case, we can bound the other three terms, so this dot product is in $[-O(\epsilon), O(\epsilon)]$. This satisfies the second condition in Matrix Class 1. □

To prove Theorem 1 for Matrix Class 1, we start with some technical lemmas.

Lemma 6. *Let $F \in \mathbb{R}_{\geq 0}^{N \times N}$ be a symmetric matrix. Let $\lambda_{\max}$ be the largest eigenvalue of F . Assuming $N \geq 2$, we have that*

$$\lambda_{\max} \geq \min_{i \neq j} F[i, j].$$

Proof. Since F is symmetric, $\lambda_{\max}$ is real and

$$\lambda_{\max} = \max_{u \neq 0} \frac{u^\top F u}{u^\top u}.$$

Let u be the all 1's vector, then

$$\begin{aligned} \lambda_{\max} &\geq \frac{1}{N} \sum_{i=j} F[i, j] \\ &\geq \frac{1}{N} \sum_{i \neq j} F[i, j] \\ &\geq \frac{1}{N} \cdot N(N-1) \min_{i \neq j} F[i, j] \\ &\geq \min_{i \neq j} F[i, j], \end{aligned}$$

where the second step follows from all the diagonal entries are non-negative, the last step follows from $N \geq 2$ □

The above implies the following result:

Corollary 7. *Let $F \in \mathbb{R}_{\geq 0}^{N \times N}$ be a block diagonal matrix. Let r be the number of $m \times m$ blocks in F for some $m \geq 2$. The $\lambda_r(F)$ is at least the smallest non-diagonal element in any $m \times m$ block ($m \geq 2$) in F .*

Proof. By Lemma 6, each $m \times m$ block B ($m \geq 2$) by itself has max eigenvalue at least $\min_{i \neq j \in [m]} B[i, j]$. The claim then follows from the fact that any eigenvalue of B is also an eigenvalue of F . □

We'll need the following function for our low-rank argument:

$$f_k(x) = \sum_{i=0}^k \frac{x^i}{i!}.$$

Note that $f_\infty(x) = e^x$.

Definition 1. Let $\epsilon \in (0, 1/10)$ and $L > 0$. We say a function $f : \mathbb{R} \rightarrow \mathbb{R}$ is (ϵ, L) -close to e^y if

$$|e^y - f(y)| \leq \epsilon \quad \text{for any } y \in [-L, L].$$

Lemma 8. For any $\epsilon \in (0, 1/10)$ and $L > 0$. If

$$D \geq 10(L + \log(1/\epsilon))$$

then function $f_D(y)$ is (ϵ, L) -close to e^y .

Proof. Recall the definition of function f_D ,

$$e^x = f_D(x) + \sum_{i=D+1}^{\infty} \frac{x^i}{i!},$$

It is sufficient to show that $|e^y - f(y)| < \epsilon$ if we have

$$\frac{x^{D+1}}{(D+1)!} \leq \frac{\epsilon}{2},$$

We can show that

$$\begin{aligned} \frac{y^D}{D!} &\leq \frac{L^D}{D!} \\ &\leq \frac{L^D}{(D/4)^D} \\ &= \left(\frac{4L}{D}\right)^D \\ &\leq (1/2)^D \\ &\leq \epsilon/10 \end{aligned}$$

where the first step follows from $|y| \leq L$, the second step follows $n! \geq (n/4)^n$, the forth step follows from $D \geq 10L$, the last step follows $D \geq 10 \log(1/\epsilon)$ and $\epsilon \in (0, 1/10)$. □

We'll also use the following fact:

Lemma 9. For any $D = o(\log n / \log d)$, we have

$$\text{rank}(f_D) \leq n^{o(1)}.$$

Proof. We can upper bound $\text{rank}(f_D(A))$ in the following sense:

$$\begin{aligned} \text{rank}(f_D(A)) &\leq (\text{rank}(A))^D \\ &\leq d^D \\ &= 2^{D \cdot \log d} \\ &= 2^{o(\log n)} \\ &= n^{o(1)}. \end{aligned}$$

where the second step follows from $\text{rank}(A) \leq d$, the forth step follows from $D = o(\frac{\log n}{\log d})$. □

Finally we're ready to prove the theorem:

Proof. The basic idea is: (i) Use $f_{k*}(b \cdot A)$ to get the low-rank approximation (ii) Use $\exp(b \cdot H)$ to get the sparse part.